Subject Description Form

Subject Code	**ITC3201T**
Subject Title	**Knitted Fabric Technology**
Credit Value	**3**
Level	**3**
Pre-requisite(s) / <Co-requisite> / (Exclusion)	ITC2202T Foundations of Textiles I
Objectives	The subject aims to provide a fundamental and comprehensive knowledge of knitted fabric technology as required by a textile technologist.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. understand basic loop formation principles using both weft knitting and warp knitting technologies; b. understand and describe different needle selection methods commonly used by flat knitting and circular machines; c. identify knitted structures formed with basic stitches and determine their knitting processes and main characteristics; d. understand and describe different shape knitting methods and perform shape knitting calculation; e. use hand flat knitting machine, circular knitting machine with multi-track cam system and electronic flat to knit fabric structures required.
Subject Synopsis/ Indicative Syllabus	**(I) Flat Knitting** Classification of flat knitting machines. Cam adjustment on a hand machine. Fabric structures formed with basic stitches and their main characteristics. Loop transfer. Knitting to shape. Introduction to electronic flat knitting machines. **(II) Circular Knitting** Classification of circular knitting machines. Needle selection principle using multi-track cam systems. Common and special knitted fabric structures produced with circular knitting machines. Positive yarn feeding system. Fabric take-down system. **(III) Warp Knitting** Classification of warp knitting machines. Loop formation principle. Representation of a warp knitted fabric structure. Primary warp knitted structures. Warp knitted fabrics from full and part set of yarn guide bars. Laying-in technique. Double warp knitting process.

Teaching/Learning Methodology	Teaching will be primarily conducted in lectures. Appropriate demonstration of knitted fabric samples, their production machineries and related tools and instruments will be conducted in the knitting laboratory.
Assessment Methods in Alignment with Intended Learning Outcomes	(see table below)

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
		a	b	c	d	e	
Continuous assessment	**50%**	✓	✓	✓	✓	✓	
1. Test	*25%*	✓	✓	✓	✓		
2. Report	*25%*			✓	✓	✓	
Examination	**50%**	✓	✓	✓	✓		
Total	100%						

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

This subject will provide students with lectures, tutorial and laboratory training. Coursework assessment will include assignments given during lectures and tutorial as well as laboratory reports, which cover all five learning outcomes. The final examination will be used to assess the first four learning outcomes.

Student Study Effort Expected

Class contact:	
• Lecture	26 Hrs.
• Lab	12 Hrs.
• Tutorial	1 Hr.
Other student study effort:	
• Self-study	66 Hrs.
Total student study effort	105 Hrs.

Reading List and References	<u>**Books**</u> Au, K. F. (ed.) (2010), *Advances in knitting technology*, Woodhead Publishing. Frederica, P. and Vikki, H. (2015), *The Knitting Book*, Dorling Kindersley Ltd. Iyer C, Mammel B. M., Schach W. (2004), *Circular Knitting*. Meisenbach Bamberg Publisher. Mayer, K. (2015/16), *Kettenwirk-Praxis*, CICO Engineering Co., LTD. Shaikh I. A. (2005), *Pocket Knitting Expert*. Textile training books, Textile Info Society. The Editors of Martingale (2016), *The Big Book of Knit Stitches*, Martingale Press.